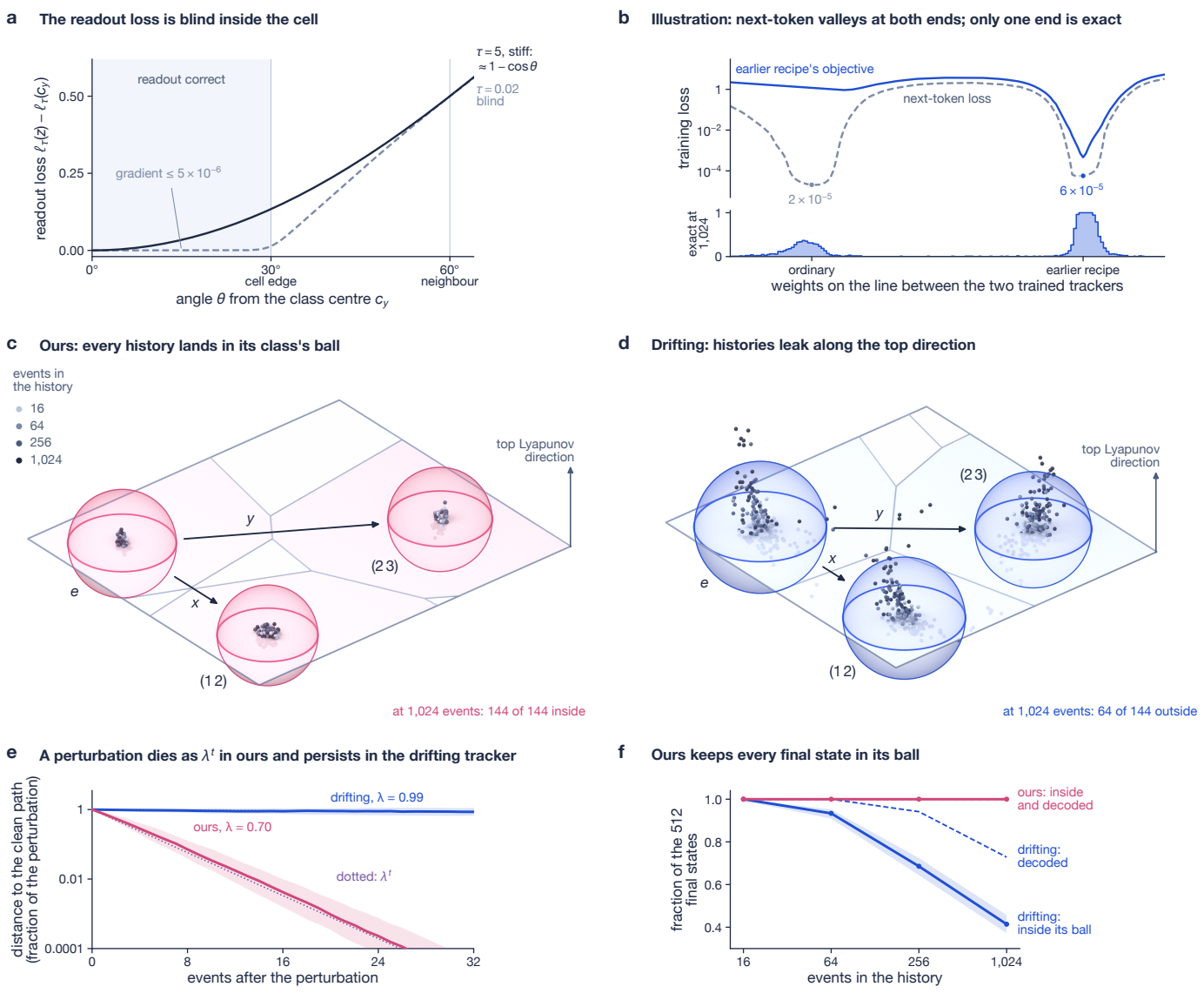

a
The readout loss is blind inside the cell
readout correct
gradient ≤ 5 × 10⁻⁶
τ = 5, stiff: ≈ 1 − cos θ
τ = 0.02 blind
readout loss ℓ_τ(z) − ℓ_τ(c_y)
0.50
0.25
0.00
0°
30°
cell edge
60°
neighbour
angle θ from the class centre c_y
b
Illustration: next-token valleys at both ends; only one end is exact
earlier recipe's objective
next-token loss
training loss
1
10⁻²
10⁻⁴
2 × 10⁻⁵
6 × 10⁻⁵
exact at 1,024
1
0
ordinary
earlier recipe
weights on the line between the two trained trackers
c
Ours: every history lands in its class's ball
events in the history
16
64
256
1,024
top Lyapunov direction
y
x
e
(2 3)
(1 2)
at 1,024 events: 144 of 144 inside
d
Drifting: histories leak along the top direction
top Lyapunov direction
(2 3)
y
x
e
(1 2)
at 1,024 events: 64 of 144 outside
e
A perturbation dies as λ^t in ours and persists in the drifting tracker
drifting, λ = 0.99
ours, λ = 0.70
dotted: λ^t
distance to the clean path (fraction of the perturbation)
1
0.01
0.0001
0
8
16
24
32
events after the perturbation
f
Ours keeps every final state in its ball
ours: inside and decoded
drifting: decoded
drifting: inside its ball
fraction of the 512 final states
1.0
0.8
0.6
0.4
16
64
256
1,024
events in the history